Work through the following exercises while the TA circulates. When you have completed the exercises, spend the rest of the laboratory working on the homework assignment.

Warm up. Solve the following equations.

- $x^2 - 5x + 6 = 0$.
- $3x^2 - x - 2 = 0$.
- $x^2 - 9 = 0$.
- $x^2 - 2x - 1 = 0$.
- $x^2 - 5x + 12 = 0$.

Exercise 1.

The absolute value function is defined as

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x \leq 0. \end{cases}$$

The function $f(x)$ is defined as the larger of x and $2 - x$.

1. Sketch the graph of $|x|$.
2. Give the domain of $f(x)$.
3. Give the range of $f(x)$.
4. Sketch the graph of $f(x)$.
5. Write $f(x)$ as a piecewise function.
6. Write $f(x)$ using the absolute value function.

Exercise 2.

Consider the function

$$g(x) = \frac{6x^2 - 7x + 2}{2x - 1}.$$

1. Give the domain of $g(x)$.
2. Give the range of $g(x)$.
3. Sketch the graph of $g(x)$.